The Climate Cost of Crowding: A Decomposition of Demography's Role in Gloabl GHG Emissions

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Abstract

Across the entire modern history, the global population has experienced significant changes, which influence sustainability substantially. Hence, achieving empirical clarity of the dominant factors in population change can offer humankind more precise and more efficient solutions for global green transition. Based on the STIRPAT model, and a transnational panel data from 1971 to 2023 covering 196 countries, this research decomposes complex demographic changes into three categories: scale, structure, and distribution, and examined their impact on global GHG emission. The results reveals that: (1) modern demographic changes increase global GHG emission; (2) instead of scale or age structure, spatial distribution is the most influencial factor of GHG; (3) urban constrcution and renewal, pollution tranfer, public service provision and digitalization mediate this effect differently; (4) heterogeneity exists in terms of aging phases, affluence and gender equality. After analytical discussions, we established a theoretical linkage between economic inclusiveness and sustainability. This research provides several insights for reference: in the domestic perspective, amending and strengthening rural and suburban areas benefits green transition,

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correcting excessive regional imbalance is essential to achieve sustainable development; in the international perspective, a stronger North-South coordinating mechanism including environment transfer payment and green technology sharing is helpful in achieving the SDGs.

Keywords: sustainability, population growth, population aging, urbanization, inclusiveness

1 Introduction

Look at what we have done: heatwaves, droughts, floods, wild fires and storms…… there are 467 pathways by which human society has been recently impacted by climate hazards caused by GHG emissions(Mora et al., 2018). The population exposed to extreme heat by the end of the century will be 4 to 8 times larger than in the 2010s(Wang et al., 2020). The dominant economic trajectory of modern history has been the global expansion of the industrialism development model. While this process has generated unprecedented material prosperity, it has also led to a substantial increase in greenhouse gas (GHG) emissions and other pollutants, as shown in Figure 1. The industrialization of the global economy has significantly contributed to global warming, sharply increasing the frequency of extreme weather events. These changes have accelerated biodiversity loss, caused severe economic damage, reduced habitable land area, and triggered localized food crises with global risk, directly threatening human life(Elahi et al., 2022). Meanwhile, the accompanying large-scale pollution has led to ecological degradation, chronic poisoning, and widespread sub-health issues. From the perspective of Buckley (1967)'s social systems theory, the structural deterioration of the environment disrupts the system's ability to process external information and energy, thereby significantly increasing economic uncertainty and the likelihood of political instability. Consequently, the need for effective global climate governance has become increasingly urgent.

But how did we bring this about? Population is always the driving force be-

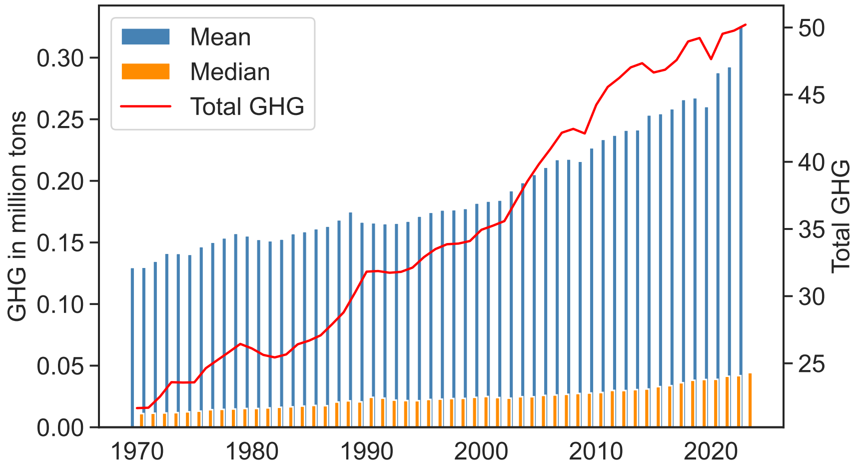
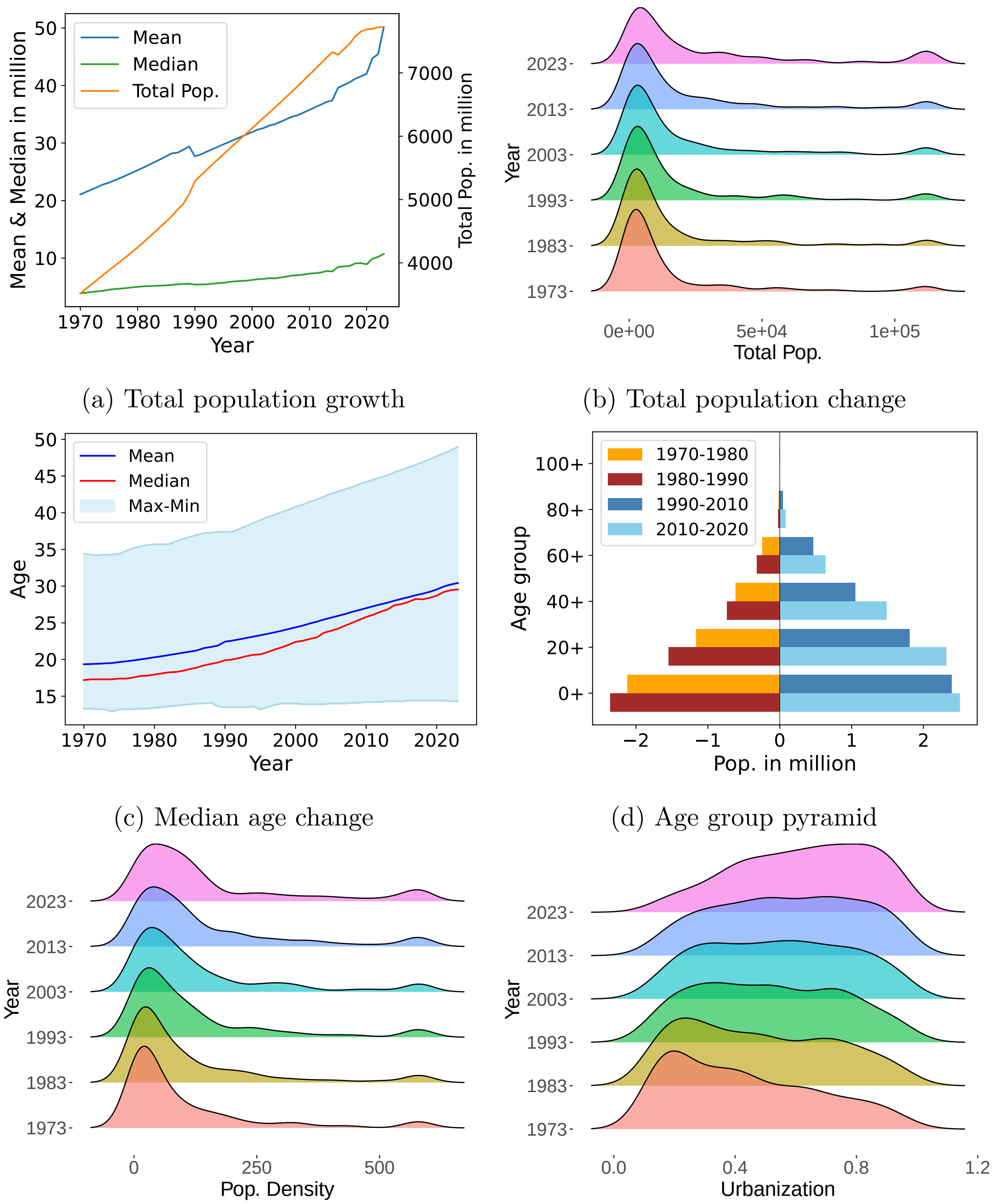


Figure 1: Global GHG emission

hind the scenes. As shown in Figure 2, modern population changes are marked by pronounced multidimensionality and imbalance. Over the past 50 years, the world's population has nearly doubled(2a). More people means higher demand, then larger-scale industrial production became necessary. Improvements in living conditions and medical care have also led to an increase in the world's median age(2c). Longer life spans naturally expand the overall market's energy demand. As more people enter working age(2d), traditional sectors cannot provide sufficient job opportunities, forcing our economies to rely increasingly on the expansion of the industrial sector to create employment. Urbanization rates are also rising(2f), meaning that the energy-intensive population is expanding. As people continue to migrate to urban areas(2e), this in itself drives further industrial expansion, increased infrastructure development, and higher energy consumption. In summary, modern population changes are driving a significant increase in GHG emissions overall. According to the classical IPAT framework in environmental economics, the two traditional drivers of climate governance—economic growth and technological advancement—are facing fundamental challenges. In recent years, global economic growth has been sluggish, and the expansion of consumption exhibits ambiguous net effects on GHG emissions. On the technological front, most countries are experiencing stagnation in research and development. Rising uncertainty has constrained investments in innovation and hindered knowledge sharing. The overall pace of scientific and technological progress is slowing, while the "digital paradox" raises doubts about the effectiveness of digital technologies in reducing emissions. Against this backdrop, the residential sector contributes 27% of global energy consumption and 17% CO_2 emissions(Nejat et al., 2015), revisiting the long-underestimated role of population dynamics may offer a new breakthrough for climate policy.



(a) Total population growth
(b) Total population change
(c) Median age change
(d) Age group pyramid
(e) Population density change
(f) Urbanization ratio change

Figure 2: Modern demographic changes

The OECD economies in Europe may serve as an insightful case, as they have simultaneously experienced population decline, aging, deurbanization, and significant reductions in GHG emissions. By the 21st century, 36% of remote areas and 13% of large cities within OECD had experienced population decline(OECD, 2023). Among European member states, Italy, Portugal, Finland, and Greece have the biggest elderly populations, at approximately 23%-24%. By 2026, the proportion of the 65+ population in Lithuania and Greece is projected to reach around 35%(OECD, 2025). Decline in the labor force has forced these developed countries to relocate their industrial sectors. Aging reduces individual energy consumption, and the exodus of people from metropolitans saves energy consumption in other aspects of urban life at the cost of increased transportation sector emissions. Additionally, once industrial upgrading is complete, a shrinking and aging population may help expand income shares to enhance economic welfare, thereby creating conditions for clean energy substitution and green technology innovation in these developed economies. Ultimately, the OECD-European economies have achieved significant success in reducing GHG emissions, as shown in Figure 3. This case study serves as a reminder to pay attention to the dynamic interaction between demographic characteristics and environmental impacts.

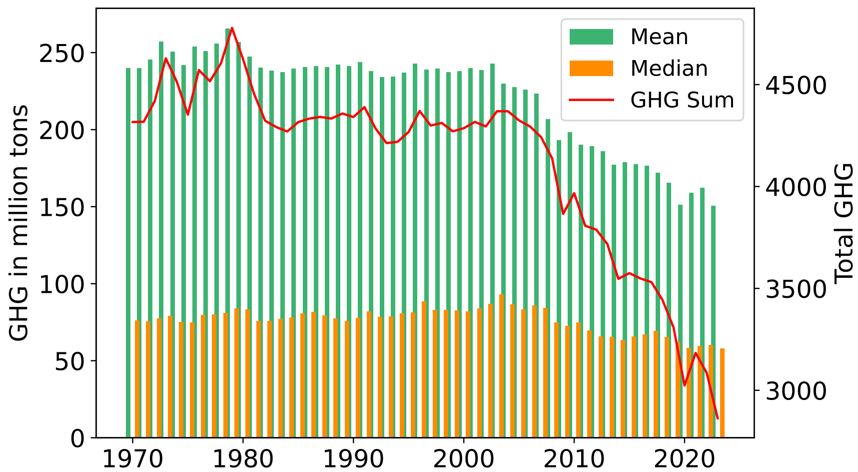


Figure 3: OECD-Europe GHG emission

Knowledge about the complex impact of comprehensive modern demographic changes on global GHG emissions is still ambiguous. Phenomena such as population decline and aging are primarily localized in a few advanced economies, with no clear global trend, and their impacts on industrial production are complex and bidirectional. Current research still lacks a systematic understanding of these mechanisms. Furthermore, in practice, limited policy resources constrain most governments, making it difficult to intervene evenly across all influencing factors. Identifying population factors with both significant impacts on GHG emissions and strong policy operability is crucial for designing effective population policies. Moreover, conventional policy tools have yet to produce effective responses to population decline and aging, highlighting the urgency of exploring alternative structural demographic variables and their policy transmission channels.

This research is aimed at estimating the systematic impact and multidimensional mechanisms of modern population changes on global GHG emissions, and identifying the most influencial demographic factor on GHG. Based on an international data set covering 196 economies across 1971 to 2023, this paper constructs an expanded STIRPAT model with more complex population factors including scale, age structure and spatial distribution. A series of mutually complementary effect decomposing methods are adopted to acquire relative importance under different focus and perspectives, including Shapley value, Forecast Error Variance Decompose (FEVD) and Random Forest variable importance ranking. Adequate robustness tests are conducted to ensure the credibility of estimations. We use the Structural Equations Model (SEM) with both the Iacobucci et al. (2007) improved Baron-Kenny method(I-BK) and the ZLC(Zhao et al., 2010) mediation test to identify the acting mechanisms, and grouped regressions with Fisher cross-group difference test for heterogeneity. Lastly, we discusses the joint significance of variables, and the role of economic inclusiveness in the green transition. Several insightful discoveries are acquired via these works. The analytical framework of this study is shown in Figure 4.

This paper finds that modern demographic changes have increased the overall GHG emission. Among three interested factors, population density—a long-

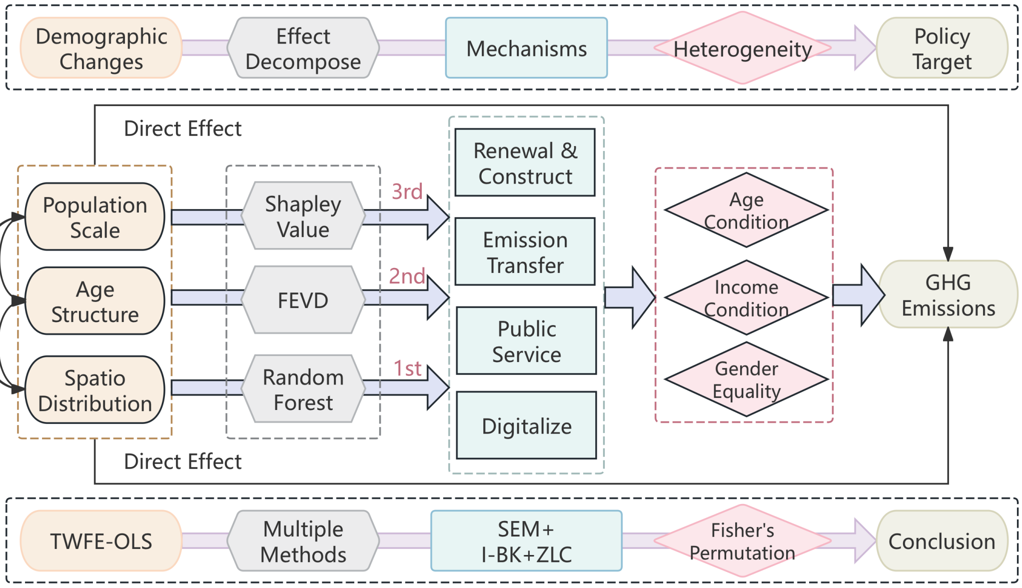


Figure 4: Analytical framework

neglected factor—shows the highest explanatory power for both the static differences and dynamic variation in global GHG emissions. Each demographic factor has different acting mechanisms. Figure 5 displays detailed logical directed acyclic graphs (DAGs) of the causality in this research. The arcs in the DAGs represent moderation effect, red arrows are direct effects, solid arrows mean positive effect, while dash arrows stand for negative effect. We can see that population growth contributes to a more attractive FDI host market, demands for more public service provision, and facilitates digitalization. Fixed assets investment in urban construction and infrastructure renewal produces emissions, net inflow FDI contains inevitable pollution transfer. On the contrary, public service provision mitigates GHG emissions by replacing individual energy consumption with public goods and service, digitalization also promote emission mitigation via multiple ways. Similar effects are observed in population aging and agglomeration. Complex heterogeneity was detected, the effect of age structure changes is more significant in older, wealthier and genderly equal economies, while the effect of distribution changes is more significant in less wealthy and genderly unequal economies.

We endeavor to provide the following possible contributions. First, this study

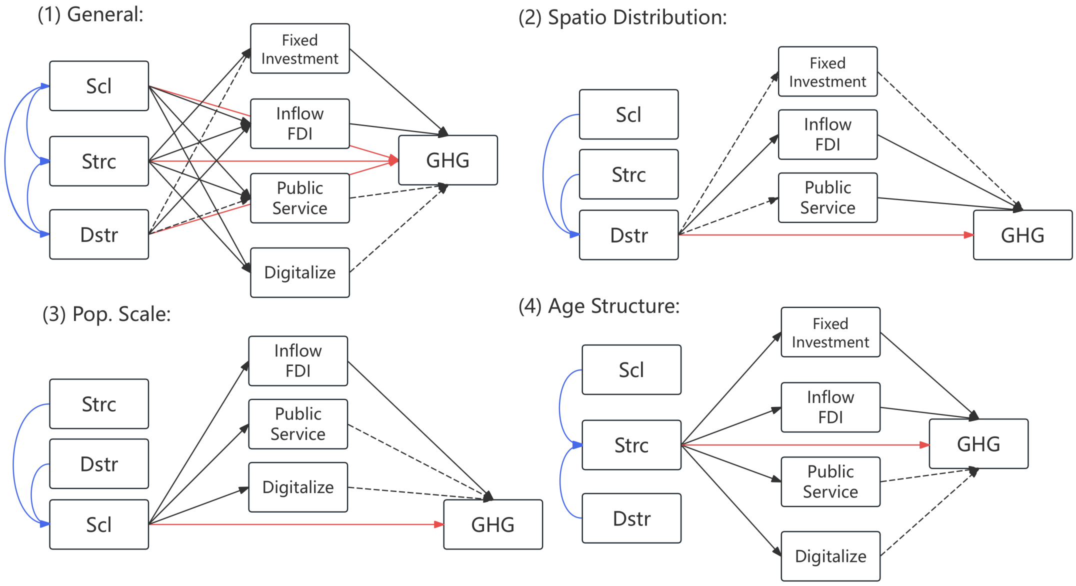


Figure 5: DAGs of the logic chain

supplements the existing researches about the nexus between demographic factors and environment impacts with long-term international evidence. Second, via multiple decomposition methods, we offer a reletively reliable conclusion supporting that spatial agglomeration in metropolitans is the most important one. Third, a complex mechanism paradigm is discovered, offering a theoretically grounded and policy-relevant foundation to achieve harmonious coexistence between humans and nature. The remaining sections are arranged as follows: Section 2 presents a brief but detailed review of related literature. Section 3 shows the theoretical analysis and research hypotheses. Section 4 describes the research design, variables, and models. Section 5 presents and analysis the results of the empirical tests. After a discussion in Sector 6, Section 7 provides conlcusion-based policy recommendations and prospects.

2 Literature review

2.1 Population growth and GHG

As the most fundamental unit of energy consumption, population growth seems intuitively correlated with increased energy consumption and, consequently, higher GHG emissions, a relationship that is relatively straightforward to verify (Magazzino et al., 2024; Khan et al., 2021; Pan et al., 2021). However, deeper research reveals more nuanced details. Firstly, a slowdown in population growth rate primarily manifests as a shrinking in household size, which actually increases residential energy consumption, thereby boosting total GHG emissions (Zhu and Peng, 2012). Secondly, the impact of population size change on GHG emissions may depend not only on the magnitude of change but also on the pre-existing population base (Ribeiro et al., 2019). Additionally, imbalances in population scale and dietary structures lead to disparities in land-use GHG emissions (Hong et al., 2021), and there exists a conflict between food production and environmental protection (Hasegawa et al., 2018).

In summary, the positive impact of population growth on GHG emissions is essentially an academic consensus. Moreover, population growth also indirectly increases GHGs and constrains emission reductions through mechanisms like the food crisis.

2.2 Population age structure and GHG

The impact of population ageing on GHG emissions currently appears inconclusive. Research from the demand side finds that older populations may have higher consumption propensities and, due to needs like heating for health and comfort, exhibit higher energy consumption. In the US and Australia, the per capita carbon footprint of the elderly is even twice the Western countries average (Zheng et al., 2022), significantly increasing GHG emissions. This effect exhibits regional heterogeneity, being insignificant in rural areas (Fan et al., 2021). Empirical evidence from China shows that changes in age structure contributed to 2.468% of the increment in carbon dioxide emissions (Pan et al., 2021); newer empirical evidence from Russia indicates that increase in the population aged 65 and above negatively affects environmental development in Russia (Ntom Udemba et al., 2024), supporting this viewpoint.

Research from the supply side argues that population ageing can reduce employment rates and labor productivity, thereby creating concomitant emission reductions due to slowed economic production (Maestas et al., 2023; Zhu and Peng, 2012). A healthier mechanism is that labor pressure caused by ageing may force enterprises to innovate their corporate governance, facilitating green innovation (Sheng et al., 2024) and providing impetus for emission reduction and long-term green growth. Furthermore, population ageing exhibits a threshold effect; after crossing the threshold value, it likely helps decouple economic growth from carbon emissions and energy consumption from carbon emissions.

It is evident that population ageing may have multi-dimensional and dialectical effects on GHGs, spanning both short and long terms. Despite the richness in detail, the average treatment effect of population ageing on the economy-wide GHG emissions has not been fully estimated.

2.3 Population spatial distribution and GHG

Cities generate three-quarters of global GHGs (Mi et al., 2019), such effect stems partly from the agglomeration of population in cities and partly from the lifestyle in cities.

People dwell densely in cities, giving cities much higher population density. A study based on Asian populous countries shows that population density leads to deteriorating environmental quality (Rahman, 2017). A more detailed study on Bangladesh indicates that population density and urbanization exacerbate carbon reduction pressure (Rahman and Alam, 2021). Within cities, ground hardening and prior localized emissions create the urban heat island effect (Huang et al., 2019), significantly amplifying energy consumption, particularly for cooling, leading to higher GHG emissions. From a dynamic perspective, one study categorized urban expansion into edge expansion and outlying expansion, finding that edge expansion significantly increases pressure on energy consumption (He et al., 2018).

Urbanization signifies a more modernized and industrialized lifestyle. Recent research by Lyubich (2025) published in the *American Economic Review* found that the place of residence contributes up to 23% to the heterogeneity in individual carbon emission behavior. Merely moving from a "low-emission neighborhood" to a "high-emission neighborhood" can increase a resident's carbon emissions from energy and commuting by up to 40%. Relocation from rural to urban areas typically accompanies changes in employment structure, while changes in industrial structure and transportation demands jointly transmit the increasing pressure of rising urban population density on GHG emissions (Yi et al., 2022). Changes in employment structure transmitted 1.28% of China's carbon emission increment (Pan et al., 2021). At least in China, urbanization implies a reduction in household size, represented by the fragmentation of larger families, which increases GHG emissions by boosting household energy consumption (Zhu and Peng, 2012).

Evidently, assuming consistent economic levels, increases in population density and urbanization are two sides of one coin, exerting pressure on GHG emissions through intricate details.

2.4 Literature synthesis and research gap

Existing literature has provided a thorough understanding of the impact of three major demographic factors — population size, age structure, and spatial distribution — on GHG emissions. Population growth often directly leads to an increase in total GHG emissions and simultaneously influences multiple factors affecting energy consumption and emissions, creating a complex mechanism of action. Population aging, on the one hand, increases the proportion of high-energy-consuming populations, while on the other hand, it reduces total emissions by lowering labor participation rates and productivity. The overall effect of these factors remains unclear. Urbanization and increased population density are often accompanied by higher GHG emissions due to energy-intensive lifestyles and more industrialized employment structures.

There are still areas for improvement in the current body of knowledge. First, the majority of literature focuses on the impact of individual demographic factors on GHG emissions, with limited research comprehensively assessing the combined effects of complex demographic changes on GHG emissions. Second, many studies use single-country data with unit root tests and Granger causality analysis, or analyze data at the provincial or municipal level within a country. Given the significant differences in population dynamics and GHG situations across countries, the findings from these studies may not effectively assess the dynamic impact of modern population changes on global GHG emissions, necessitating supplementary cross-country panel regression analysis. Finally, due to the narrow scope of domestic data and the monotonously use of simplified decomposition methods, such as the logarithmic mean Divisia index (LMDI)(Alajmi, 2021), the few studies that have addressed effect decomposition have not reached consistent conclusions.

Given the limitations of the existing literature, this study simultaneously considers the most representative population characteristics in terms of scale, age structure, and spatial distribution to examine the impact of composite population changes on global GHG emissions. It analyzes data from 196 countries spanning 50 years from 1971 to 2023 and employs multiple complementary effect decomposition methods for estimation.

3 Theoretical analysis

3.1 Direct effect

The IPAT and its derivative STIRPAT models are distilled from a large body of important research on climate impacts from the early 21st century(York et al., 2003), they affirm the influence of demographic factors, particularly population size, on climate impacts. Population growth implies higher energy consumption from both the demand and supply sides, thereby leading to higher GHG emissions(Magazzino et al., 2024). As an increasing proportion of the world's population enters working age, the size of the labor force is directly correlated with industrial scale(Maestas et al., 2023). The entry and exit of populations into and out of working age thus exert synchronous effects on industrial GHG emissions. Population aging reduces industrial GHG output by lowering labor participation rates, but the elderly population has a higher tendency toward energy consumption(Zheng et al., 2022). Considering that aging primarily occurs in developed economies, whose industrial structures do not heavily rely on manufacturing, the consumption growth effect of aging outweighs the labor reduction effect, thereby expanding GHG emissions overall. The increase in global average population density in modern history has primarily occurred in urban areas. This process is always accompanied by a reduction in household size, improved living standards, and changes in energy consumption habits(Rahman and Alam, 2021). The industrialization of urban population has significantly increased global GHG emissions by expanding the total energy consumption. Generally speaking, modern population change characterized by high growth rate, larger working population and urbanization increases global GHG emissions. Research hypotheses are proposed below.

H1. Modern population changes increase GHG emissions.

H1a. Ceteris paribus, population growth causes GHG emission increasement.

H1b. Ceteris paribus, working population ratio is positively correlated with GHG emissions.

H1c. Ceteris paribus, high density population agglomeration correlates with high emissions.

3.2 Mediating effect

3.2.1 Urbanization, fixed assets investment and GHG

Housing quality constitutes a crucial mediating factor in energy poverty alleviating and subjective well-being enhancement (Nie et al., 2021). This suggests that urbanization invariably demands greater construction land and more comfortable housing, an effect amplified by urban population growth and rising population density. Especially in resource-based cities, the interaction between urban population growth and fixed asset investment has significantly driven the expansion of construction land in these cities (Li et al., 2022). Both urbanization and aging also heighten the demand for urban public infrastructure. An increase in the urban elderly population underscores the importance of accessible facilities, nursing homes, hospitals, parks, etc., and boosts the demand for public transportation (Zhang and Xu, 2022). This entails substantial fixed asset investment, primarily in new construction and renovation/retrofitting, whether from governmental or private sectors. The construction industry is a major emitter of atmospheric particulate matter and GHGs, and its upstream and downstream industries – including mining, mineral processing, manufacturing, and transportation – also generate significant GHG emissions. Empirical studies have confirmed a negative correlation between real estate investment and air quality (Chen and Lee, 2020). Therefore, the agglomeration of population in urban areas and the rising proportion of the elderly population may lead to higher local GHG emissions by stimulating fixed asset investment in public infrastructure.

Hypothesis 2. Modern demographic changes influence domestic GHG emissions by reshaping the structure and scale of fixed asset investment.

3.2.2 Labor, FDI inflow and GHG

Population is the most fundamental component of the consumer market. An increase in the scale of population with purchasing power may enhance the attractiveness of FDI and promote OFDI through stronger economic growth rates.

In this dynamic, manufacturing FDI may become a vehicle for transnational pollution transfer, i.e., the "pollution haven" hypothesis. Many studies focusing on developing economies have found that FDI flowing into Latin America has led to an increase in host country pollution levels(Sapkota and Bastola, 2017). For representative developing countries in other continents (Mexico, Indonesia, Nigeria, Turkey), FDI exhibits an inverted U-shaped relationship with ecological footprints(Balsalobre-Lorente et al., 2019). Additionally, FDI flowing into China has been shown to facilitate its urbanization(Wu and Chen, 2016), thereby generating more GHG emissions from urban construction and daily life.

FDI also has a dialectical impact on GHG emissions. High-quality FDI can mitigate pollution transfer by exporting technology(Ntom Udemba et al., 2024), and capital-exporting countries often have more advanced environmental protection concepts, which can help host countries achieve environmentally friendly development through foreign investment(Zhang and Zhou, 2016). The impact of FDI on GHG emissions may also be influenced by heterogeneity of economic stages. A study based on quantile regression indicates that FDI at higher quantiles helps reduce carbon emissions(Zhu et al., 2016). However, given that green industries and environment friendly development remain rare in developing countries, and the majority of international investment still focuses on low-end manufacturing, the overall impact of FDI on GHG emissions is more likely to be positively correlated.

Hypothesis 3. Modern demographic changes affect GHG emissions through their impact on the composition of FDI.

3.2.3 Public service provision, government expenditure and GHG

The increase in population size, urban population density, and the number of school-age and elderly residents all necessitate more adequate, comprehensive, and accessible public service provision. When population inflows occur, the development of social infrastructure becomes a decisive factor in urban vitality(Lan et al., 2020), and the demand for public services in areas such as education, healthcare, elderly care, and public transportation will continue to grow alongside population growth, aging, and urbanization(Zhang and Xu, 2022). Additionally, labor concentration and maturation in cities imply more complex urban economic activities and higher network centrality, necessitating increased government investment to improve market mechanisms and provide positive externalities. On the other hand, the scale effect of population agglemeration might lower the marginal cost of public service provision, therefore save government expenditure.

New structural economics argues that the role of a rational government should be to improve and supplement the market system, providing subsidies and coordination for positive externalities and public goods (such as industry experience, R&D trial and error, and social responsibility) that cannot be formed due to the private sector's inability to bear the costs, thereby enhancing the overall efficiency of the economy(Lin, 2011). Although governance costs and government financial activities also generate certain carbon dioxide emissions(Le and Ozturk, 2020), government spending focused on public goods supply and systematically considering and absorbing externalities costs makes the same project investments "greener" compared to the private sector. Additionally, by improving overall economic efficiency, carbon intensity is reduced, thereby decreasing total GHG emissions.

Hypothesis 4. Modern demographic changes promote the enhancement of governmental governance capacity, thereby strengthening the regulation and control of GHG emissions.

3.2.4 Access, digitalization and GHG

The widespread adoption of digital technology is closely tied to demographic characteristics. In terms of hardware, the construction of digital infrastructure such as signal towers, communication base stations, and fiber-optic cables involves high upfront fixed costs but low marginal costs, meaning that cost-sharing and return on investment are only achievable in areas with high user density. In terms of applications, small businesses face low economies of scale when incurring costs for digitalization, and population dispersion and low accessibility lead to inefficient e-commerce delivery. Therefore, population growth and urban agglomeration can provide a favorable environment for digital development. At the same time, the elderly population generally has lower digital skills and weaker willingness and ability to use digital technologies, which may delay the widespread adoption of digitalization.

The role of digitalization in green transformation and sustainable development has been well-established. The improvement of digital infrastructure can promote carbon neutrality by enhancing carbon emission efficiency, innovation in carbon-intensive industries, and carbon-neutral technology innovation(Zhou et al., 2024). Meanwhile, digital applications such as FinTech can promote innovation in carbon-intensive industries(Zhou et al., 2025), thereby fostering growth of green economy(Awais et al., 2023). Mobile money is also effective in enhancing financial accessibility to improve residential welfare(Munyegera and Matsumoto, 2016). Furthermore, the green effect of digitalization demonstrates significant spatial spillover effects(Ren et al., 2023).

Hypothesis 5. Modern demographic changes facilitate digitalization, which helps optimize economic processes and reduce the GHG intensity.

4 Research design

4.1 Model construction

4.1.1 Baseline model

The IPAT model offers a powerful framework for analyzing ecological change and its driving factors. It decomposes environmental impact into three key socio-economic determinants: population characteristics, affluence, and technology, thereby establishing a foundation for examining the link between demographic variables and environmental change. Building upon the traditional IPAT framework, the STIRPAT model introduces a stochastic term to account for unobserved factors and measurement errors, and it allows for variable elasticities, providing greater flexibility in fitting empirical data. Based on the log-linear STIRPAT model discussed before, this study decomposes the population factor (P) into three characteristic variables: absolute scale, age structure, and spatial distribution. In Equation (1), A_{it}, T_{it}, and $\mathbf{X}_{it}$ represent affluence, technology, and other factors, respectively, all included as control variables. To account for country-specific characteristics and temporal dynamics, a two-way fixed effects panel model is employed.

$$\begin{aligned} lnGHG_{it} =& \alpha_0 + \beta_1 lnScl_{it} + \beta_2 lnStrc_{it} + \beta_3 lnDstr_{it} \\ & + \gamma_1 lnA_{it} + \gamma_2 lnT_{it} + \boldsymbol{\gamma}'\mathbf{X}_{it} + \lambda_i + \delta_t + \varepsilon_{it} \end{aligned} \tag{1}$$

In Equation (1), β_1, β_2, and β_3 are three coefficients of interest. Their statistical significance indicates the correlation between different population characteristics and GHG emissions. Moreover, as standardized coefficients are used in this study, their relative magnitudes reflect, to some extent, the relative importance of each component within P. λ_i denotes individual fixed effects, δ_t represents time fixed effects, and ϵ_{it} is the random error term.

4.1.2 FEVD model

Given that the baseline regression model and linear decomposition methods are sensitive to linearity assumptions, this study further employs the FEVD method based on a vector auto-regression (VAR) model to obtain more robust effect decomposition. The VAR model does not rely on predefined causal directions, allows for high-dimensional interactions among variables, and its multi-period autoregressive structure better captures the dynamic explanatory power of independent variables on the dependent variable. Following the approach of Lyubich (2025), the variance decomposition model is specified as shown in Equation (2).

$$
\begin{aligned}
Var(GHG_{it}) = {} & Var(Scl_{it}\beta_1) + Var(Strc_{it}\beta_2) + Var(Dstr_{it}\beta_3) + Var(X_{it}\beta_4) \\
& + 2Cov(Scl_{it}\beta_1, Strc_{it}\beta_2) + 2Cov(Scl_{it}\beta_1, Dstr_{it}\beta_3) \\
& + 2Cov(Strc_{it}\beta_2, Dstr_{it}\beta_3) \\
& + 2Cov(Scl_{it}\beta_1, X_{it}\beta_4) + 2Cov(Strc_{it}\beta_2, X_{it}\beta_4) \\
& + 2Cov(Dstr_{it}\beta_3, X_{it}\beta_4) + Var(\varepsilon_{it})
\end{aligned}
\tag{2}
$$

4.1.3 Mechanism SEM model

Traditional mediation models based on linear regression are prone to collinearity and endogeneity issues, after a bulk of Monte Carlo simulations, Iacobucci et al. (2007) confirmed that structural equation modeling (SEM) is a more appropriate approach for examing complex pathways(Fan et al., 2016). Therefore, this paper combines the SEM with the improved Iacobucci-Baron Kenny mediation test to simultaneously examine the factor loadings and indirect effects of the four channels and calculates the proportion of indirect effects in the total effect. We also adopts the ZLC method developed by Zhao et al. (2010) to indentify the mediation type.

$$
\begin{cases}
Mediator_{it}^{n} = \gamma_0 + \gamma_1 lnP_{it} + \gamma_2 lnA_{it} + \gamma_3 lnT_{it} + \boldsymbol{\gamma}'\mathbf{X}_{it} + \zeta_{it}^{1} \\
lnGHG_{it} = \beta_0 + \beta_1 Mediator_{it}^{n} + \beta_2 lnP_{it} + \beta_3 lnA_{it} + \beta_4 lnT_{it} + \boldsymbol{\gamma}''\mathbf{X}_{it} + \zeta_{it}^{2}
\end{cases}
\tag{3}
$$

In Equation (3), four mediators are estimated seperately, P_{it} denotes the three population variables, their coefficients γ_1 and the mediators' coefficients β_1 are estimated simultaneously to tell whether the mediation effect exists.

4.1.4 Mutual mediation SEM model

Since three independent variables are included simultaneously, their mutual interaction should receive investigation. Severe collinearity could occur when one of the variables is partly determined by another. Also, the effect decomposition result would be less reliable if joint effect is neglected. Given significant causality is essential in mediation effect, this paper adopts a mediation SEM similar to the mechanism model to detect wheter there's causality problem in the baseline model. In Equation (4), each demographic variables are taken as a mediator of the other two, if the path load coefficients γ and effect coefficients β are both insignificant, then there's no worry about mutual mediation.

$$\begin{cases} Scl_{it} = \alpha_1 + \gamma_1 Strc_{it} + \gamma_2 Dstr_{it} + \boldsymbol{\gamma}^1 \mathbf{X}_{it} + \zeta_{it}^1 \\ Strc_{it} = \alpha_2 + \gamma_3 Scl_{it} + \gamma_4 Dstr_{it} + \boldsymbol{\gamma}^2 \mathbf{X}_{it} + \zeta_{it}^2 \\ Dstr_{it} = \alpha_3 + \gamma_5 Scl_{it} + \gamma_6 Strc_{it} + \boldsymbol{\gamma}^3 \mathbf{X}_{it} + \zeta_{it}^3 \\ GHG_{it} = \alpha_0 + \beta_1 Scl_{it} + \beta_2 Strc_{it} + \beta_3 Dstr_{it} + \boldsymbol{\gamma}^0 \mathbf{X}_{it} + \zeta_{it}^0 \end{cases} \tag{4}$$

4.1.5 Dynamic auto-correlation model

GHG emissions of an economies is related with its industrial structure and energy mix, which is rather stable over short periods. This means that GHG of the present period is very likely to be affected by GHG from the previous periods. Thus, this paper adopts a dynamic auto-correlation model to ensure such phenomenon doesn't cause severe causal inference mistakes.

$$\begin{aligned} lnGHG_{it} =& \alpha_0 + \alpha_1 lnGHG_{i,t-1} + \beta_1 lnScl_{it} + \beta_2 lnStrc_{it} + \beta_3 lnDstr_{it} \\ &+ \gamma_1 lnA_{it} + \gamma_2 lnT_{it} + \boldsymbol{\gamma}' \mathbf{X}_{it} + \lambda_i + \delta_t + \varepsilon_{it} \end{aligned} \tag{5}$$

4.2 Variable definitions and data sources

As the dependent variable of this study, GHG emissions are measured in terms of the carbon dioxide equivalent of various greenhouse gases(European Commission. Joint Research Centre. and IEA., 2024). Among the three population characteristic variables, the population growth rate is used to capture the dynamic change in population scale and its relationship with GHG emissions. For age structure, the working-age population ratio is selected, as changes in the labor force (aged

15 to 60) are most directly linked to economic production. Given that population agglomeration in urban and rural areas may have divergent effects on GHG emissions, in order to avoid potential biases from urban-rural dichotomies, population density is used to represent the spatial distribution of the population.

Among the control variables specified by the IPAT model, per capita consumption expenditure is used to represent affluence, following common practice. Since the focus is on the environmental effects of technology, carbon efficiency (CO_2 emissions per unit of economic output) is employed as a proxy for technological level. Additional control variables include population aging, urbanization, and economic openness. The elderly population (aged 60+) is normally considered lack of productive capacity and exhibits distinct consumption preferences, thereby influencing GHG emissions through both production and consumption channels. Urbanization increases local energy consumption through improved living standards, but may simultaneously enhance resource efficiency via agglomeration, thus exerting a dual effect on emissions. Economic openness is measured by total imports, reflecting the extent to which physical production is outsourced abroad, thereby reducing pollution generated domestically.

Four mechanism variables are selected based on the theoretical hypotheses, with detailed descriptions provided in Table 1.

The data used in this study are drawn from authoritative and reliable sources. GHG emissions and carbon efficiency are obtained from the Emissions Database for Global Atmospheric Research (EDGAR), most population data are sourced from the United Nations World Population Prospects (WPP) 2024. Data on population distribution and urbanization are taken from the United Nations World Urbanization Prospects (WUP) 2018. Economic data involving monetary values primarily come from the World Bank's World Development Indicators (WDI), with the remaining economic variables sourced from the Global Macro Database (GMD), compiled by research teams at the National University of Singapore and the University of California, Berkeley. EDGAR is an internationally recognized

Table 1: Variable definition & data source

Variable	Symbol	Definition	Source
Greenhouse gas emissions	*GHG*	Annual total emissions of greenhouse gases	EDGAR
Population scale change	*Scl*	Annual growth rate of total population	UN WPP
Population age structure	*Strc*	Share of working-age population in total population	UN WPP
Population spatial distribution	*Dstr*	Population per square kilometer	UN WUP
Affluence	*cons*	Annual per capita total consumption expenditure	GMD
Technological level	*ci*	Carbon efficiency (e.g., GHG emissions per unit of output or energy)	EDGAR
Population aging	*old*	Share of population aged 65 and above	UN WPP
Urbanization	*urbr*	Share of population living in urban areas	UN WUP
Economic openness	*imp*	Total imports	GMD
Urban renewal & construction	*Fx_Ivst*	Gross fixed assets investment as a percentage of GDP	GMD
Potential pollution transfer	*nFDI*	Net foreign direct investment	WB WDI
Public service provision	*PSP*	Government expenditure as a percentage of GDP	GMD
Digitalization	*Digit*	Internet penetration rate	WB WDI

independent database under the European Commission. The GMD is a unified, open-source dataset that integrates historical and contemporary data from 110 sources, providing annual macroeconomic indicators for 243 countries from early records through projections to 2030(Müller et al., 2025).

4.3 Descriptive statistics

After merging the datasets, minimal additional processing was conducted. Invalid samples with consecutive years of missing data were removed, resulting in a final dataset comprising complete observations for 196 countries from 1971 to 2023. All continuous variables were transformed using natural logarithms to align with the log-linear model specification. Additionally, Z-score standardization was applied to allow for meaningful comparison of coefficient magnitudes. Descriptive statistics are presented in Table 2.

Table 2: Descriptive statistics

		Original				Z-Score	
Variable	Obs	Mean	SD	Min	Max	Min	Max
GHG	9796	9.785	2.526	1.504	16.585	-3.279	2.692
Scl	9796	1.692	1.980	-71.060	36.090	-36.742	17.372
Strc	9796	-0.519	0.128	-1.792	-0.159	-9.980	2.826
Dstr	9796	4.050	1.572	-2.303	10.013	-4.041	3.792
cons	9796	4.365	0.265	2.179	5.698	-8.260	5.038
ci	9796	-5.131	3.538	-17.404	21.419	-3.469	7.505
old	9796	0.064	0.045	0.000	0.259	-1.404	4.320
urbr	9796	-079	0.595	-3.559	0.156	-4.658	1.591
imp	9796	3.583	0.705	-18.473	6.437	-31.271	4.046
Fx_Ivst	9692	4.808	0.076	4.581	5.552	-2.989	9.789
nFDI	8832	0.544	2.851	-35.933	73.383	-12.793	25.546
PSP	8116	4.830	0.106	4.605	8.144	-2.118	31.224
Digit	5355	1.995	2.605	-10.953	4.605	-4.9699	1.0017

5 Empirical analysis

5.1 Baseline regression

The results of the baseline regression are presented in Table 3. Columns (1) and (2) include only the three explanatory variables of interest, while Columns (3) and (4) incorporate the control variables specified by the STIRPAT model. The final two columns include the full set of control variables. Different combinations of fixed effects are compared to ensure the robustness of the results. The findings indicate that all three population characteristic variables exhibit a significantly positive association with GHG emissions at the 95% confidence level or higher across all specifications. This provides preliminary support for Hypothesis 1, suggesting that contemporary demographic changes, on the whole, exert upward pressure on GHG emissions.

Table 3: Baseline regression

	GHG					
	(1)	(2)	(3)	(4)	(5)	(6)
Scl	0.0123^{**}	0.0125^{**}	0.0083^{*}	0.0086^{**}	0.0087^{**}	0.0093^{**}
	(0.0059)	(0.0059)	(0.0043)	(0.0042)	(0.0040)	(0.0042)
Strc	0.0617^{***}	0.0588^{***}	0.0404^{**}	0.0385^{**}	0.0390^{**}	0.0409^{**}
	(0.0168)	(0.0168)	(0.0170)	(0.0170)	(0.0167)	(0.0177)
Dstr	0.5694^{***}	0.5487^{***}	0.5620^{***}	0.5486^{***}	0.5291^{***}	0.5719^{***}
	(0.0548)	(0.0750)	(0.0446)	(0.0654)	(0.0532)	(0.0762)
cons			-0.0355^{***}	-0.0349^{***}	-0.0358^{***}	-0.0345^{***}
			(0.0123)	(0.0123)	(0.0123)	(0.0121)
ci			-0.1509^{***}	-0.1507^{***}	-0.1453^{***}	-0.1476^{***}
			(0.0540)	(0.0549)	(0.0544)	(0.0559)
old					0.0109	0.0304
					(0.0169)	(0.0277)
urbr					0.0246	0.0323
					(0.0231)	(0.0256)
imp					0.0054	0.0054
					(0.0037)	(0.0038)
Constant	0.0000^{***}	-0.0119	0.0000^{***}	-0.0046	0.0000^{***}	0.0346
	(0.0000)	(0.0228)	(0.0000)	(0.0213)	(0.0000)	(0.0330)
Controls	N	N	M	M	Y	Y
Year FE	N	Y	N	Y	N	Y
Country FE	Y	Y	Y	Y	Y	Y
r2	0.6249	0.6279	0.6694	0.6714	0.6718	0.6746
N	9796	9796	9796	9796	9796	9796

Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Y indicates the corresponding factor is included. N denotes exclusion.
M: control variables specified by the STIRPAT model (A, T) only.

5.2 Effect decomposition

5.2.1 Standardized coefficients comparison

Standardized coefficients from the multiple linear regression represent the average marginal effect of each independent variable on the dependent variable, controlling for other factors. Among the three population variables, population density exhibits the largest and most statistically significant standardized coefficient, indicating that, ceteris paribus, it has the strongest explanatory power for variations in GHG emissions. This comparison of standardized coefficients supports the view that population density should be a primary consideration in green governance strategies involving demographic factors.

5.2.2 Shapley decomposition

In a multiple linear regression model, the coefficient of an independent variable reflects its marginal effect on the dependent variable, holding all other factors constant. However, as linear models are susceptible to interference from confounding factors, comparing marginal effects alone may be insufficient for robust inference. Therefore, this study further employs Shapley decomposition—a widely recognized technique for assessing the relative contribution of multiple variables to model performance. Specifically, the Shapley method systematically removes each explanatory variable from the baseline model, records the resulting change in the R^2, and averages these effects across all possible variable combinations. The resulting R^2 variations reflect each variable's explanatory power, or relative importance, in the model.

Table 4 presents the results of the Shapley decomposition, where column numbers correspond to those in Table 3. The findings indicate that the spatial distribution of the population is the most influential factor, accounting for over 70% of the explained variance. This result is consistent with the comparison of standardized coefficients.

Table 4: Shapley decomposition result

		Shapley Value			% in P		
		(1)	(3)	(5)	(1)	(3)	(5)
Scl	d_*Scl*	0.0033	0.0028	0.0022	0.52	0.49	0.38
Strc	d_*Strc*	0.1947	0.163	0.1320	31.15	29.04	22.57
	d_*old*			0.0130			2.22
Dstr	d_*Dstr*	0.4270	0.4036	0.3099	68.32	70.47	52.99
	d_*urbr*			0.1277			21.84
A	d_*cons*		0.0226	0.0218			
	d_*imp*			0.0033			
T	d_*ci*		0.0741	0.0619			
Total		0.06249	0.6694	0.6718	100	100	100

It is worth noting that the Shapley method does not support factor variables, and thus cannot directly control for individual and year fixed effects. To address this limitation, all variables were demeaned within groups prior to the Shapley decomposition, removing year fixed effect. While demeaning prevents comparisons across groups, this does not compromise the analysis, as the study focuses on the relative contribution of population-related factors to GHG emissions rather than cross-group differences. Therefore, no valuable information is lost in this process.

5.2.3 FEVD

Next, from a dynamic perspective, this paper conducted a VAR-based Forecast Error Variance Decomposition(FEVD). Both standardized coefficient comparison and Shapley decomposition focus on static evaluation, yet FEVD provides a way to measure each variable's contribution to the deviation of the dependent variable. Similar to a recent research from Lyubich (2025), we set *GHG* as the response variable, and the three components of demographic shift are positioned as impulse variables seperately. Table 5 presents the detailed results, which includes 10 future steps of *GHG* and share of correpsonding impulses. The results from the FEVD model provided similar conclusion compared with the Shapley method and baseline regression coefficients, spatial distribution remains the prime factor in complex demographic change influencing global GHG emission.

Table 5: FEVD result

Step	2	3	4	5	6	7	8	9	10	Avg.
Scl	7.59	16.89	19.50	22.04	22.94	23.42	23.34	22.99	22.42	18.11
Strc	0.12	0.39	1.78	4.40	7.81	11.87	16.24	20.77	25.29	8.87
Dstr	92.30	82.72	78.73	73.56	69.25	64.71	60.42	56.24	52.28	63.02

5.2.4 Random Forest ranking

Lastly, we conducted an variable importance ranking using Random Forest machine learning algorithm. Random Forest is an ensemble learning method that builds a large number of decision trees and aggregates their predictions to improve model stability and predictive accuracy. Within this framework, variable importance is assessed using the percentage increase in mean squared error (%IncMSE), which captures each variable's marginal contribution by comparing prediction errors before and after permutation. Compared with Shapley decomposition and traditional variance decomposition, Random Forest does not rely on linear assumptions and can effectively capture nonlinear relationships and interaction effects among variables. Moreover, while variance decomposition typically requires the underlying series to be stationary, Random Forest imposes no such constraint, making it more suitable for empirical settings involving trends or structural shifts. In our analysis, the importance ranking obtained from Random Forest is consistent with the results from both Shapley and variance-based approaches, jointly identifying the spatial distribution of population as the most influential demographic determinant of global greenhouse gas emissions.

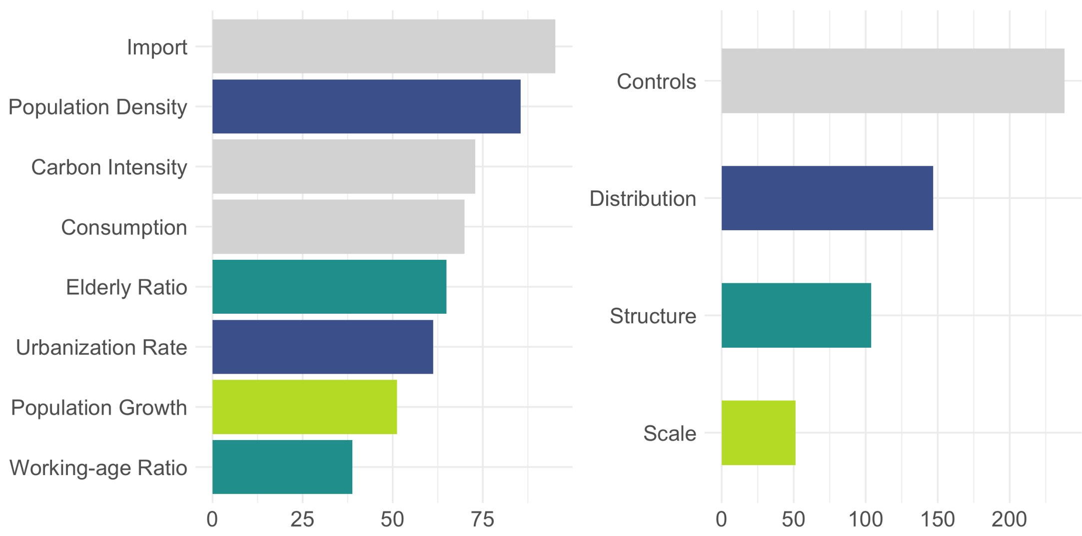


Figure 6: Random Forest variable importance ranking

5.3 Robustness tests

5.3.1 Treating extreme samples

Significant disparities in development levels and GHG emissions across countries have long been a defining and inevitable feature of the global landscape. As a result, datasets covering all available samples may be influenced by extreme observations, which can lead to overestimation of effects and reduce the representativeness of conclusions. To address this issue, all variables were winsorized by replacing the top and bottom 1% of values with the nearest observations. Table 6 presents the regression results after winsorization. The three coefficients of interest remain significantly positive, suggesting that the baseline results are robust to the influence of outliers.

Table 6: Robustness: treating extreme samples

	GHG					
	(1)	(2)	(3)	(4)	(5)	(6)
*Scl*_w	0.0313**	0.0320**	0.0228*	0.0235*	0.0246**	0.0255**
	(0.0151)	(0.0152)	(0.0122)	(0.0122)	(0.0115)	(0.0128)
*Strc*_w	0.0794***	0.0758***	0.0542***	0.0510**	0.0522***	0.0531**
	(0.0168)	(0.0181)	(0.0196)	(0.0211)	(0.0196)	(0.0215)
*Dstr*_w	0.5446***	0.5247***	0.5414***	0.5222***	0.5100***	0.5204***
	(0.0532)	(0.0727)	(0.0453)	(0.0645)	(0.0500)	(0.0765)
Controls	N	N	M	M	Y	Y
Year FE	N	Y	N	Y	N	Y
Country FE	Y	Y	Y	Y	Y	Y
r2	0.6142	0.6169	0.6522	0.6540	0.6541	0.6558
N	9796	9796	9796	9796	9796	9796

Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.3.2 Bootstrap estimation

The traditional panel OLS method relies on several strong assumptions, which are often violated in empirical research. To mitigate this risk, this study employs 300-replication bootstrap tests. The bootstrap approach constructs resampled datasets through repeated random sampling and does not depend on any specific distributional assumptions. As shown in Table 7, the results are consistent with the baseline regression, confirming the robustness of the estimated effects.

Table 7: Robustness: Bootstrap estimation

	GHG					
	(1)	(2)	(3)	(4)	(5)	(6)
Scl	0.0123***	0.0125***	0.0083***	0.0086***	0.0087***	0.0093***
	(0.0034)	(0.0034)	(0.0030)	(0.0030)	(0.0030)	(0.0031)
Strc	0.0617***	0.0588***	0.0404***	0.0385***	0.0390***	0.0409***
	(0.0042)	(0.0044)	(0.0039)	(0.0040)	(0.0039)	(0.0043)
Dstr	0.5694***	0.5487***	0.5620***	0.5486***	0.5291***	0.5719***
	(0.0107)	(0.0142)	(0.0089)	(0.0129)	(0.0109)	(0.0159)
Controls	N	N	M	M	Y	Y
Year FE	N	Y	N	Y	N	Y
Country FE	Y	Y	Y	Y	Y	Y
r2	0.6249	0.6279	0.6694	0.6714	0.6718	0.6746
N	9796	9796	9796	9796	9796	9796

Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.3.3 Mutual effect

Since the empirical model in this paper includes three key explanatory variables simultaneously, significant correlations among them may lead to multicollinearity, which could adversely affect the precision of the estimates and the stability of coefficient interpretation. In addition, if these variables are highly correlated, the effect decomposition results discussed earlier may become ambiguous due to overlapping causal pathways. Therefore, this section constructs a structural equation model and applies dual mediation tests to examine whether the three variables serve as mutual mediators. As shown in Table 12, there is no significant linear mediation effect among the three explanatory variables. This suggests that reciprocal causality remains within an acceptable range, or that such effects do not significantly influence GHG estimation. Overall, we find no evidence of serious multicollinearity in the model.

Table 8: SEM mutual mediation test

X	(M <- X)	(Y <- M <- X)	Z-Value	RIT	Type
	M = *Scl*				
Strc	-0.0856	-0.008	-0.012	-	No mediation
	(7.0281)	(0.626)			
Dstr	-0.0452	-0.004	-0.012	-	No mediation
	(3.6966)	(0.329)			
	M = *Strc*				
Scl	-0.0550	-0.011	-0.012	-	No mediation
	(4.6710)	(0.932)			
Dstr	0.0765	0.015	0.024	-	No mediation
	(3.1660)	(0.632)			
	M = *Dstr*				
Scl	-0.0087	0.001	0.002	-	No mediation
	(4.3763)	(0.411)			
Strc	0.2095	-0.020	-0.037	-	No mediation
	(4.3763)	(0.530)			

5.3.4 Bilateral causality

Strictly speaking, the OLS method identifies correlations rather than causal relationships. Bidirectional causality is one of the main sources of endogeneity. To address this, the temporal order of variables is leveraged. Table 9 presents the results using lagged independent variables: columns 1–3 use one-year lags, while columns 4–6 use two-year lags. The lagged coefficients remain significantly positive, suggesting that the previously estimated effects persist over time. Since the dependent variable cannot influence past values of the independent variables, these findings help to mitigate reverse causality and provide endogeneity-robust evidence for the direction of causality.

5.3.5 Dynamic auto-correlation correction

The GHG emission of an economy is largely influenced by long-term structural factors, including industrial structure, energy structure, and emission habits. Moreover, the urgency of emission reduction is often linked to the GHG stock from previous periods. Therefore, current GHG emissions are likely affected by those

Table 9: Endogeneity treatment: bilateral causality

	GHG					
	(1)	(2)	(3)	(4)	(5)	(6)
L.*Scl*	0.0146^{**}	0.0108^{**}	0.0112^{**}	0.0160^{**}	0.0123^{**}	0.0127^{***}
	(0.0061)	(0.0045)	(0.0045)	(0.0063)	(0.0048)	(0.0048)
L.*Strc*	0.0581^{***}	0.0383^{**}	0.0407^{**}	0.0571^{***}	0.0381^{**}	0.0406^{**}
	(0.0164)	(0.0165)	(0.0172)	(0.0158)	(0.0160)	(0.0166)
L.*Dstr*	0.5462^{***}	0.5462^{***}	0.5595^{***}	0.5429^{***}	0.5433^{***}	0.5466^{***}
	(0.0746)	(0.0646)	(0.0744)	(0.0741)	(0.0639)	(0.0727)
Lag	I	I	I	II	II	II
Controls	N	M	Y	N	M	Y
FEs	Y	Y	Y	Y	Y	Y
r2	0.6263	0.6695	0.6726	0.6238	0.6657	0.6690
N	9599	9599	9599	9402	9402	9402

Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

of previous periods, reflecting dynamic autocorrelation in the model. Such autocorrelation can lead to endogeneity, which requires bias-resistant estimation using flexible instrumental variables via Generalized Method of Moments (GMM). Table 10 presents the relevant test results. The insignificance of the AR(2) statistic suggests that the current IV set is not correlated with the error, satisfying validity. The Hansen overidentification test is also insignificant, supporting the exogeneity of the IV set and the overall model specification. After controlling first-order dynamic autocorrelation, the effects of the three population factors on GHG remain consistent with the baseline regression. Even the relative importance of the variables is consistent, with the spatial distribution of population still being the most influential factor.

Table 10: Endogeneity treatment: dynamic auto-correlation GMM estimation

GHG	Coef.	Std. Err.	z	P>|z|	[95% conf.	interval]
L.*GHG*	0.771	0.046	16.58	0.000	0.680	0.862
Scl	0.005	0.001	3.66	0.000	0.002	0.007
Strc	0.017	0.009	1.91	0.057	-0.000	0.034
Dstr	0.142	0.040	3.58	0.000	0.064	0.220
Arellano-Bond test for AR(2) :				P > |z| = 0.243		
Hansen test of overid. restrictions:				P > chi2 = 0.999		

5.4 Mechanism tests

Based on the theoretical framework, this section conducts a mechanism analysis focusing on investment structure, including urban renewal and construction(Fx_Ivst), international pollution transfer($nFDI$), public service provision(PSP), and digitalization($Digit$). The results are presented in Table 11.

Table 11: Mediation-SEM result

X	(M <- X)	(Y <- M <- X)	Z-Value	RIT	Type
	M = *Fx_Ivst*				
Scl	-0.0211	-0.001	-1.671	-	Direct-only
	(0.0112)	(0.000)			
Strc	0.0921***	0.003***	3.500	1.8%	Complementary
	(0.0140)	(0.001)			
Dstr	-0.0946***	-0.003***	-3.783	3.6%	Complementary
	(0.0104)	(0.001)			
	M = *nFDI*				
Scl	0.0496***	0.008***	3.907	9.9%	Complementary
	(0.0124)	(0.002)			
Strc	0.0641***	0.010***	3.785	6.4%	Complementary
	(0.0166)	(0.003)			
Dstr	0.0419***	0.007***	3.564	6.6%	Competitive
	(0.0115)	(0.002)			
	M = *PSP*				
Scl	0.0436**	-0.004**	-3.031	4.4%	Competitive
	(0.0137)	(0.001)			
Strc	0.0686***	-0.006***	-3.691	3.6%	Competitive
	(0.0174)	(0.002)			
Dstr	-0.1595***	0.015***	8.444	27.2%	Competitive
	(0.0119)	(0.002)			
	M = *Digit*				
Scl	0.0821***	-0.005**	-3.387	4.7%	Competitive
	(0.0191)	(0.001)			
Strc	0.4347***	-0.028***	-5.505	25.6%	Competitive
	(0.0193)	(0.005)			
Dstr	-0.0203	0.001	1.504	-	Direct-only
	(0.0127)	(0.001)			

Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The results reveal a set of complex and multidimensional mediation pathways. In general, fixed assets investment in urban renewal and construction is positively correlated with GHG emissions, net inflow FDI serves as a carrier of international emission transfer, which is coherent with the ”Pollution Paradise” hypothesis. Public service provision and digitalization mitigate GHG emission via the economy of scale and dematerialization, respectively. In detail, the mediation network is consistent with theoretical derivations:

1. Population growth significantly increases both public service provision and facilitates internet penetration, which together help offset the GHG emissions driven by population expansion.

2. A higher proportion of the working-age population stimulates production-oriented fixed asset investment, and attracts greater FDI inflows, thereby increases GHG emissions. Simultaneously, a larger working population broadens the tax base, strengthens the government’s environmental governance capacity, and promotes digital transformation, partially mitigating the emission growth.

3. Increased population density tightens land availability, supresses urban renewal, it thus limits emissions from construction sector. High density also leads to a more attractive international investment host market, such impetus missmatch combined with crowding-out effects, high population dnesity increases GHG emissions from the private sector.

These findings not only illustrate the diverse mechanisms through which contemporary demographic shifts influence GHG emissions, but also yield several key implications. First, investment structure and its externalities are crucial in determining environmental outcomes: while production-focused investment tends to raise emissions, spending in areas such as digitalization, elderly care, and green infrastructure tends to reduce them. Second, international investment serves as a channel for both pollution relocation and the cross-border transmission of demographic shocks. Finally, the consistent negative correlation between internet penetration and GHG emissions suggests that most economies remain on the left side of the "digital U-curve," indicating considerable untapped potential for digitalization-driven emission reductions.

5.5 Heterogeneity test

Population change may have different characteristics depending on a country's age structure. Countries with different demographic foundations often have distinct economic systems and historical backgrounds, which may lead to heterogeneous effects on GHG emissions. High-income and low-income countries also differ in industrial composition and population dynamics, resulting in varied transmission channels for population impacts. Furthermore, economic inclusiveness affects fertility intentions, labor participation, and population mobility, all of which shape the link between population change and environmental outcomes.

Based on these considerations, this study explores heterogeneity from three perspectives: age structure (measured by median age), income level (GNI per capita), and inclusiveness (gender parity in school enrollment). For each indicator, countries are divided into two groups using the median of standardized values to ensure balanced sample sizes. Coefficient differences between groups are tested. Figure 7 displays the results of the subgroup regressions, detailed analytical statistics are reported in the appendix Table 13.

In Figure 7, the vertical axes stand for the coefficient number, while different colorbar visualizes coefficients from different subgroups, 95% confidential intervals are added.

According to the results of the coefficient difference tests, the impact of population growth on GHG emissions does not show significant heterogeneity across groups. As the core driver of economic production, the relationship between population growth and GHG emissions appears to be universal and consistent.

In contrast, age structure shows notable heterogeneous effects. In countries with a higher median age or higher income levels, an increase in the working-age population tends to raise GHG emissions. However, in countries with a younger

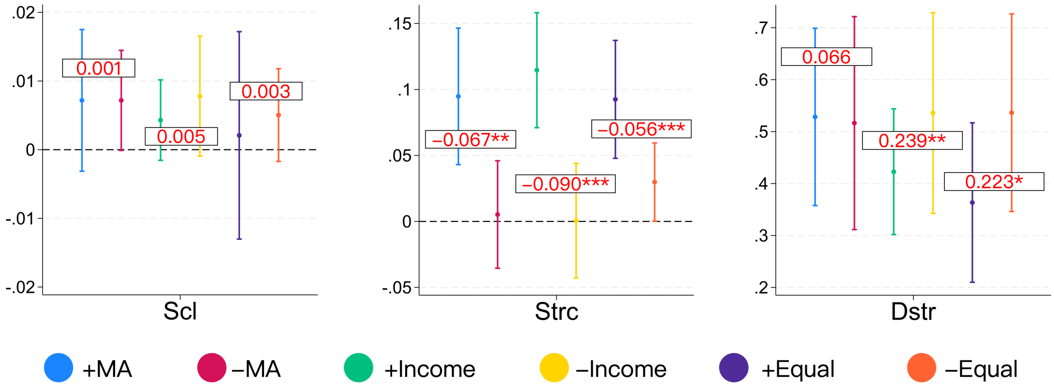


Figure 7: Fisher coss-group coefficient difference test

Number in frames denotes the cross-group difference, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. For color-challenged readers, the legend is in the same order as the figure.

population or lower income, no similar evidence is found. This divergence is largely due to the fact that in more industrialized and economically advanced countries, people entering the working-age range are more likely to engage in actual production, thereby generating more emissions. In less developed countries with weaker industrial bases, labor tends to be absorbed by low-emission traditional sectors. Additionally, working-age individuals are the primary consumers. In high-income countries, higher living standards lead to greater indirect emissions, while in low-income nations, limited consumption results in lower GHG output. Gender equality also plays a role: in more equal societies, women are more likely to participate in the labor force, contributing to both economic output and increased emissions.

Spatial concentration of population also produces heterogeneous effects. In low- and middle-income countries, denser populations are often associated with higher GHG emissions due to inadequate public infrastructure, weak energy governance, and inefficient production management. In contrast, high-income countries may offset such effects through better environmental governance and public service provision. Gender equality again exerts influence: in less equal countries, population clustering is linked to higher emissions, suggesting that gender parity reflects broader aspects of institutional maturity and collective environmental awareness—factors that are essential for sustainable urban development.

The heterogeneity analysis suggests that differences in population structure, economic development levels, and social inclusiveness significantly shape how demographic changes impact the environment. Therefore, ecological governance policies should be context-specific, taking into account each country's unique conditions in terms of demographic dividends, industrial structure, and governance capacity to formulate more targeted green development strategies.

5.6 Joint effects

Although we have ruled out the possibility of mutual mediation in our robustness tests, we do not rule out the existence of some conditional relationship among the three population variables. For example, population growth can sometimes be viewed as the primary driver of urban expansion(Mahtta et al., 2022). To this end, we constructed a moderation effect model to test for joint significance among the variables. The regression results are presented in Table 12.

Table 12: Joint effect test result

	GHG			
	(1)	(2)	(3)	(4)
Scl#Strc	-0.0079**			-0.0035
	(0.0036)			(0.0031)
Scl#Dstr		-0.0137***		-0.0127***
		(0.0049)		(0.0039)
Strc#Dstr			0.0323***	0.0315***
			(0.0105)	(0.0102)
Scl#Strc#Dstr				-0.0011
				(0.0030)
Scl	0.0078*	0.0116***	0.0093**	0.0108***
	(0.0041)	(0.0043)	(0.0043)	(0.0038)
Strc	0.0430**	0.0427**	0.0471***	0.0492***
	(0.0171)	(0.0181)	(0.0156)	(0.0148)
Dstr	0.5842***	0.5915***	0.5776***	0.6014***
	(0.0755)	(0.0747)	(0.0828)	(0.0798)
Controls	Y	Y	Y	Y
FEs	Y	Y	Y	Y
r2	0.6762	0.6778	0.6917	0.6949
N	9796	9796	9796	9796

Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The results reveal an interesting set of moderation effects:

1. Working-age population negatively moderates the impact of population growth on GHG emissions. This may be because the younger population has stronger green technology absorption capacity and policy responsiveness, higher GHG elasticity mitigates the environmental negative impacts of population growth.

2. Population density negatively moderates the impact of population growth on GHG emissions. High-density areas typically have more developed infrastructure, public transportation, centralized heating, and other economies of scale facilities, resulting in lower marginal emissions per additional population. For example, in densely populated areas, new residents often use subways instead of private cars, live in high-rise buildings or slums instead of dispersed single-family homes, thus have a lower average carbon footprint.

3. Population density positively moderates the impact of age structure on GHG emissions, as this interaction term implies the high-density agglomeration of working population, which often entails large-scale industrial activities and more severe urban issues.

In summary, age structure and distribution moderate the marginal effects of population growth on GHG emissions to some extent. Increased density and optimized labor force structure can help mitigate the emissions pressure caused by population growth. However, when high-density areas concentrate large labor populations, emissions may intensify, and structural pollution risks should be monitored.

6 Discussions: Inclusiveness and sustainability

What insights can we gain from this study? The growing and maturing population in modern history has produced a large amount of GHG, with population density playing the most critical role. The expansion of population scale, the increase in the proportion of working population, the advancement of urbanization, and the rapid growth of greenhouse gas emissions together form a complete dynamic landscape of a developing economy: A person is often born in a rural area or suburb, lack of economic prospects urges them to separate from their families and migrate to cities in search of work when reaching adulthood. Some find employment in office buildings with 24/7 lighting and air conditioning, while others work in factories with round-the-clock shifts. Greenhouse gases are emitted alongside their sweat, and the fact is that they were never given a choice. The midnight streets of Dhaka, Dongguan, Doha, and Bangalore bear witness to their stories.

Economic inclusiveness has become a focal point in developed economies, with the fair sharing of development outcomes and equal participation widely recognized as critical to enhancing residents' welfare and long-term growth. However, in many developing economies, significant wealth gaps and regional imbalances remain the norm. Following the theoretical tradition of neo-Marxist spatial critique(Buckley, 1967), we conceptualize inclusiveness as a synthesis of "spatial inclusiveness" and "vertical inclusiveness".

6.1 Spatial inclusiveness

The "center-periphery" regional development pattern was highly conducive to fostering spatial growth poles in the early stages, but as the economy enters a consolidation phase, promoting more balanced regional development takes on new significance in the contemporary era. On the one hand, a more balanced spatial structure helps reduce population density in central metropolises, thereby reversing the causal pathway where population density drives GHG emissions. More importantly, regional balance may provide conditions for the restoration of family structures. The multi-center urbanization pattern has fragmented families located in transitional zones, forcing family members to live separately in multiple small, high-energy-consuming living units. This "atomized" family structure and its association with consumerism have been extensively discussed in broader sociological contexts. Through regional balanced development, providing equal and accessible economic opportunities may alleviate overall GHG emissions by restoring the efficiency of family aggregation.

Additionally, the outflow of population from small towns and rural areas to large cities has led to a spatial mismatch between the supply and demand of ecological services, reducing the efficiency of ecological service value development and utilization: urban residents must increase energy consumption to offset ecological deficits, while rural residents' per capita ecological resources remain underutilized or underdeveloped. Dispersing the population and fostering small and medium-sized towns to a limited extent can promote the coupling of population and ecological carrying capacity, providing conditions for a new type of "ecological urbanization" that organically integrates towns with the natural environment.

6.2 Vertical inclusiveness

The standards for evaluating officials' and governments' performance should not be mechanical GDP growth rates or other economic indicators, but should instead be directly linked to residents' well-being. This may be the key to ending the "necessary evil" of economic inequality in the "post-growth era". Accessibility to ecological services should be considered an important dimension of well-being. Currently, the ecological economics community generally acknowledges that residents have the same demand for ecological products and services as they do for other goods and services. However, urban management has yet to systematically incorporate these into the public service framework, failing to provide ecological products or services in the same manner as public services such as affordable housing and social safety nets. We believe this aligns with the goal of equitable sharing of economic benefits. Currently, the wealthy often have more green spaces, better air quality, and more accessible natural resources, while common high-density residential areas may not provide sufficient green space to meet ecological needs, and slums are even nearly impossible to be associated with ecological services. We propose that, while ensuring basic living conditions, more proactive efforts should be made to provide greening, parks, plastic walking paths, and waste collection services to middle- and low-income communities to achieve equitable access to ecological services.

Equal and effective broad social participation is also an important manifestation of vertical inclusivity. As mentioned earlier, the elderly typically have a higher carbon footprint, so their attention and participation are crucial components of the economic green transition. However, in reality, the elderly remain in a passive position in public policy-making, their voices are not heard, and due to low digital literacy, they struggle to form opinions. Therefore, digital inclusion technologies and community work mechanisms should be utilized to assist the elderly in participating in green policy design and promoting inclusive ecological governance. Similarly, our heterogeneity analysis indicates that income levels and gender equality significantly influence the impact of demographic factors on GHG emissions. Thus, traditional vulnerable groups such as low-income individuals and women must be granted equal influence in the formulation of green transition schemes.

More specific research and arguments will be conducted in our future work.

7 Conclusions & Policy implications

7.1 Research conclusions

To enhance theoretical understanding of modern population dynamics, this paper investigates the changes in three key population characteristics—scale, structure, and distribution—based on panel data from 196 countries between 1971 and 2023. It examines their impacts on global GHG emissions. Using Shapley decomposition, FEVD and Random Forest ranking, it evaluates the relative importance of each population factor, and employs I-BK SEM to reveal the complex mechanisms through which demographic changes influence GHG emissions, as well as their differentiated effects across countries with diverse demographic profiles. These findings offer practical policy implications for achieving green development through targeted population governance.

Globally, population characteristics have undergone profound transformations in the modern era. The total population has nearly doubled since the 1970s. The share of the working-age population generally increased alongside total population growth, but began to slow significantly after the 2010s, lagging behind total population growth and signaling the onset of notable population aging. Meanwhile, the upper bound of the median age has approached 50 years, while the lower bound has remained relatively stable, reflecting stark global disparities in human development. Urbanization has progressed rapidly, surpassing 50% of the world's total population by the early 21st century, which means that the majority now residing in urban areas.

These demographic shifts have generally contributed to increased GHG emissions. Controlling for other factors, rising population density is found to be the most consistent driver of increased emissions, as confirmed by a series of robustness checks. Investment structure is a criteria mechanism: fixed asset investment in urban expansion and renewal tends to increase emissions; international investment flows are frequently associated with pollution transfer; public service provision contributes positively by managing environmental externalities; and digital investment, at the current stage, can effectively reduce overall emissions. Heterogeneity analysis reveals several notable findings: demographic changes in mature economies—with higher median ages and per capita income—are more likely to result in increased emissions, which is linked to the level of industrialization. Gender equality affects emissions by influencing labor force participation. Moreover, the impact of population agglomeration on GHG emissions depends on the quality of agglomeration, with low-quality clustering (e.g., low income, high inequality) more likely to exacerbate emissions. Mild mutual moderations exist between demographic variables, making the interaction effect more complex. Lastly, spatial and vertical inclusiveness also have their role in green transition and sustainable development.

7.2 Policy implications

Achieving harmony between humanity and nature is a prerequisite for sustainable development and inclusive growth. This requires greater attention to the interplay between demographic characteristics and environmental quality. Based on the findings of this study, several policy recommendations are proposed:

1. Given the widespread challenges of population governance, population density is a relatively feasible and effective policy target. Promoting healthy declines in population density through more balanced regional development while maintaining quality of life, plays an often-overlooked but important role in enabling green transition.

2. The structure of investment should be guided toward low-carbon and digital sectors. Resources should be directed toward digital infrastructure, renewable energy, and smart manufacturing to reduce reliance on traditional emission-intensive investments. In particular, digital technologies should be further applied to improve energy efficiency and pollution monitoring.

3. Environmental regulation of international investment should be strengthened. A coordinated system of cross-border environmental standards should be established to prevent pollution "leakage" through international investment. Stricter environmental assessments should be applied to foreign investments. Moreover, upstream and downstream cooperation within global value chains should be enhanced to ensure that green technologies and governance resources are transferred alongside capital flows, with environmental externalities priced into investment decisions.

4. The environmental effectiveness of public services should be improved. Public expenditures should prioritize areas with strong positive externalities—

such as clean energy, public transportation, and ecological restoration—to drive emissions reductions of the entire economy. At the same time, ESG ratings should be widely promoted to encourage businesses to engage in similar externality governance.

5. Attention should be paid to the quality of population agglomeration. Ecological welfare is closely linked with other welfares. Policymakers should also address income inequality and gender disparities to improve the quality of demographic clustering and its environmental outcomes.

7.3 Limitations

Aiming to provide more precise knowledge about modern demographic shift and its relationship with global green transition, this paper proposes three representative population characteristics, namely scale, age structure, and spatial distribution. However, the complex demographic shift contains other less discussed but equally important factors. Also, to support quantitive research, this paper picks population growth rate, working population ratio, and population density to reflect changes of population. Future researches can apply more accurate and comprehensive indicators to capture wider demographic factors.

Appendix

For the readers's convenience, the result table for the heterogeneity test was provided below. All statistics are consistent with the visualized edition as shown in Figure 7.

Table 13: Heterogeneity test result

	GHG					
	+MA	-MA	+Income	-Income	+Equality	-Equality
Scl	0.0072	0.0072^{*}	0.0043	0.0078^{*}	0.0021	0.0050
	(0.0052)	(0.0037)	(0.0030)	(0.0044)	(0.0076)	(0.0034)
Strc	0.0948^{***}	0.0052	0.1146^{***}	0.0006	0.0925^{***}	0.0298^{**}
	(0.0261)	(0.0206)	(0.0220)	(0.0220)	(0.0226)	(0.0150)
Dstr	0.5283^{***}	0.5164^{***}	0.4228^{***}	0.5358^{***}	0.3634^{***}	0.5365^{***}
	(0.0862)	(0.1035)	(0.0611)	(0.0975)	(0.0776)	(0.0963)
	Fisher cross-group difference test					
Scl	0.001		0.005		0.003	
Strc	-0.067^{**}		-0.090^{***}		-0.059^{**}	
Dstr	0.066		0.239^{**}		0.223^{*}	
Controls	Y	Y	Y	Y	Y	Y
Country FE	Y	Y	Y	Y	Y	Y
r2	0.5815	0.7349	0.6177	0.7429	0.6629	0.7602
N	4782	4860	4413	4431	2794	2822

Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
The + and – before each grouping criterion indicate values above or below the median for that year, respectively.
MA refers to median age; Income refers to GNI per capita; Equality represents gender equality measured by the ratio of school enrollment between boys and girls.

CRediT authorship contribution statement

Lei Li: Conceptualization, Methodology, Formal analysis, Writing-Original Draft, Visualization. **Yinfeng Chen:** Writing-Review and Editing, Methodology, Data curation.

Declaration of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to polish language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data Availability Statement

Data available from the authors upon request.

Ethical approval

This article does not contain any studies with human participants or animals performed by any authors.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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